



**Generation Interconnection
System Impact Study Report
for**

Queue Project AF1-064

WESTON 69 KV

33.4 MW Capacity / 50 MW Energy

August 2020

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is ATSI.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 General

The Interconnection Customer (IC), has proposed a Solar generating facility located in Wood County, Ohio. The installed facilities will have a total capability of 50 MW with 33.4 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is March 31, 2022. This study does not imply a TO commitment to this in-service date. The interconnection customer has requested cost to accelerate the supplemental project s1953¹ in order to reduce the direction connection cost for AF1-064. FE will proceed with accelerating the supplemental project s1953 with a signed Interim ISA that is to be developed by PJM and the Interconnection Customer.

Queue Number	AF1-064
Project Name	WESTON 69 KV
State	Ohio
County	Wood
Transmission Owner	ATSI
MFO	50
MWE	50
MWC	33.4
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

¹ Supplemental Project s1953 for Richland Substation Upgrades and New Richland-Weston 69 kV line in the ATSI zone is currently scheduled for completion by 12.31.2023. The customer is requesting to accelerate the completion to May 31, 2023.

4 Point of Interconnection

AF1-064 will interconnect with the ATSI on transmission system at the Weston 69 kV substation.

The interconnection of the project at the selected POI will be accomplished by expanding Weston 69 kV Substation into a five (5) breaker ring bus substation and loop in the Bowling Green No2-Midway 69 kV Line to eliminate the three terminal line. The ring bus will provide a line exit for the AF1-064 project. The interconnection customer has requested the cost to accelerate the supplemental project s1953 in order to reduce the direct connection cost for AF1-064. FE will proceed with accelerating the supplemental project s1953 with a signed Interim ISA that is to be developed by PJM and the interconnection customer.

Attachment 1 shows a one-line diagram of the proposed primary Direct Connection facilities for the AF1-064 generation project to connect to the FirstEnergy ("FE") Transmission System. The IC will be responsible for constructing the facilities on its side of the POI, including the Attachment Facilities which connect the generator to the FE Transmission System's Direct Connection facilities.

5 Cost Summary

The AF1-064 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$4,299,400
Allocation towards System Network Upgrade Costs*	\$0 (See Note 2)
Total Costs	\$4,299,400

***As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.**

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such

as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

The interconnection of the project at the Primary POI will be accomplished by expanding Weston 69 kV Substation into a five (5) breaker ring bus substation and loop in the Bowling Green No2-Midway 69 kV Line to eliminate the three terminal line. The ring bus will provide a line exit for the AF1-064 project. The interconnection customer has requested the cost to accelerate the supplemental project s1953 in order to reduce the direct connection cost for AF1-064. FE will proceed with accelerating the supplemental project s1953 with a signed Interim ISA that is to be developed by PJM and the interconnection customer.

The total physical interconnection costs is given in the table below:

Description	Total Cost
Install SCADA and MOAB for the New Line Exit at Weston Substation	\$82,600
Provide a 69 kV Line Exit at Weston Substation for AF1-064	\$743,200
Provide assistance to the developer on the customer-owned revenue metering design, perform a remote meter checkout, and facilitate the meter's installation in the FE MV-90 system.	\$2,500
Project Management	\$284,900
Drawing review, nameplates, and CD drawing update; add to HV circuit diagram at AF1-064 Customer Sub	\$30,600
Acceleration of s1953 to accommodate AF1-064.	\$3,155,600
Total Physical Interconnection Costs	\$4,299,400

7 Schedule

Based on the scope of work for the interconnection facilities, it is expected to take a minimum of **24 months** after the signing of an Interconnection Construction Service Agreement and construction kickoff call to complete the installation. This assumes that there will be no environmental issues with any of the new properties associated with this project, that there will be no delays in acquiring the necessary permits for implementing the defined Direct Connection and network upgrades, and that all transmission system outages will be allowed when requested.

The schedule for any required Network Impact Reinforcements will be more clearly identified in future study phases. The estimated time to complete each of the required reinforcements is identified in the “System Reinforcements” section of the report.

8 Transmission Owner Analysis

FE performed an analysis of its underlying transmission <100 kV system. The AF1-064 project did not contribute to any overloads on the FE transmission < 100 kV system.

9 Interconnection Customer Requirements

9.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE’s “Requirements for Transmission Connected Facilities” document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

The IC has requested a non-standard GSU transformer winding configuration. This transformer is in violation of section 14.2.6 of FE’s “Requirements for Transmission Connected Facilities” document and will not be accepted. The GSU transformer must have a grounded wye connection on the high (utility) side and a delta connection on the low (generator) side.

9.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FE’s “Requirements for Transmission Connected Facilities” document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated 69 kV circuit breaker to protect the AF1-064 generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.

2. The purchase and installation of the minimum required FE generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the FE Transmission System Control Center.
4. Compliance with the FE and PJM generator power factor and voltage control requirements.
5. The execution of a back-up service agreement to serve the customer load supplied from the AF1-064 generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.

The IC will also be required to meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FE Transmission System.

9.3 Power Factor Requirements

The IC shall design its solar Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the FE transmission system.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Back Panel temperature (Fahrenheit)
- Irradiance (Watts/meter²)
- Ambient air temperature (Fahrenheit) – (Accepted, not required)
- Wind speed (meters/second) – (Accepted, not required)

- Wind direction (decimal degrees from true north) – (Accepted, not required)

10.3 Interconnected Transmission Owner Requirements

The IC will be required to comply with all FE revenue metering requirements for generation interconnection customers which can be found in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>

11 Summer Peak Analysis

The Queue Project AF1-064 was evaluated as a 50.0 MW (Capacity 33.4 MW) injection at the Weston 69 kV substation in the ATSI area. Project AF1-064 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-064 was studied with a commercial probability of 100%. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/D C	MW IMPACT
43720294	243009	05FRMNT	138.0	AEP	243008	05FREMT	138.0	AEP	1	AEP_P2-2_#517_05FOS TOR 345_1	bus	251.0	106.05	107.65	AC	2.76
43720647	243009	05FRMNT	138.0	AEP	243008	05FREMT	138.0	AEP	1	AEP_P4_#517_05FOS TOR 345_A1	breaker	251.0	106.05	107.65	AC	2.76

11.4 Steady-State Voltage Requirements

To be determined in Facilities Study phase.

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-064	Upgrade Number
43720647,43720294	1	05FRMNT 138.0 kV - 05FREMCT 138.0 kV Ckt 1	AEP: A sag study will be required on the 4.0 miles of ACSR ~ 795 ~ 45/7 ~ TERN - Conductor Section 1 to mitigate the overload. Depending on the sag study results, the cost for this upgrade is expected to be around \$20,000 (no remediation required, just sag study) and \$6 million (complete line reconductor/rebuild). New rating after sag study: S/N:251 S/E: 335 MVA. Time Estimate: a) Sag Study: 6-12 months. b) Rebuild: The standard time required for construction differs from state to state. An approximate construction time would be 24 to 36 months after signing an interconnection agreement. Project Type : FAC Cost : \$20,000 New Ratings : 251/335 Time Estimate : 6-12 Months	\$200,000	\$0	N6297.1
			TOTAL COST	\$200,000	\$0	

Note: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	ACID C	MW IMPACT
43720647	243009	05FRMNT	AEP	243008	05FREMC T	AEP	1	AEP_P4_#517_05FOSTOR 345_A1	breaker	251.0	106.05	107.65	AC	2.76

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
238601	02FRMENG 1	5.3360	50/50	5.3360
238602	02FRMENG 2	5.3360	50/50	5.3360
238603	02FRMENG 3	9.6331	50/50	9.6331
244357	05GRANGER EL	-0.1967	Adder	-0.23
931951	AB1-107 1	14.5816	Adder	17.15
931961	AB1-107 2	26.0727	Adder	30.67
934251	AD1-052 C1	0.3960	50/50	0.3960
934252	AD1-052 E1	0.9946	50/50	0.9946
934261	AD1-052 C2	0.3960	50/50	0.3960
934262	AD1-052 E2	0.9946	50/50	0.9946
934891	AD1-118	3.0595	Adder	3.6
938911	AE1-119	24.0388	Adder	28.28
941761	AE2-176 C	8.1465	50/50	8.1465
941762	AE2-176 E	5.4310	50/50	5.4310
943961	AF1-064 C O1	1.5683	Adder	1.85
943962	AF1-064 E O1	0.7794	Adder	0.92
NEWTON	NEWTON	0.4169	Confirmed LTF	0.4169
FARMERCITY	FARMERCITY	0.0171	Confirmed LTF	0.0171
G-007A	G-007A	0.0647	Confirmed LTF	0.0647
VFT	VFT	0.1935	Confirmed LTF	0.1935
PRAIRIE	PRAIRIE	0.9686	Confirmed LTF	0.9686
COFFEEN	COFFEEN	0.0688	Confirmed LTF	0.0688
CHEOAH	CHEOAH	0.2427	Confirmed LTF	0.2427
EDWARDS	EDWARDS	0.0770	Confirmed LTF	0.0770
TILTON	TILTON	0.2167	Confirmed LTF	0.2167
MADISON	MADISON	1.1592	Confirmed LTF	1.1592
GIBSON	GIBSON	0.2697	Confirmed LTF	0.2697
CALDERWOOD	CALDERWOOD	0.2420	Confirmed LTF	0.2420
BLUEG	BLUEG	1.0451	Confirmed LTF	1.0451
TRIMBLE	TRIMBLE	0.3378	Confirmed LTF	0.3378
CATAWBA	CATAWBA	0.1540	Confirmed LTF	0.1540
CBM-W1	CBM-W1	10.1331	Confirmed LTF	10.1331

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AB1-107	Bayshore-GM Powertrain 138 kV & Lallendorf 345kV	Engineering and Procurement
AD1-052	Freemont Energy Center	Under Construction
AD1-118	Lemoyne	Active
AE1-119	Lemoyne 345 kV	Active
AE2-176	Groton 138 kV Solar	Active
AF1-064	Weston 69 kV	Active

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
AEP_P2-2_#517_05FOSTOR 345_1	CONTINGENCY 'AEP_P2-2_#517_05FOSTOR 345_1' OPEN BRANCH FROM BUS 241901 TO BUS 242936 CKT 1 / 241901 02LALLEND
AEP_P4_#517_05FOSTOR 345_A1	CONTINGENCY 'AEP_P4_#517_05FOSTOR 345_A1' OPEN BRANCH FROM BUS 241901 TO BUS 242936 CKT 1 / 241901 02LALLEND

12 Light Load Analysis

Not required for solar projects.

13 Short Circuit Analysis

The following Breakers are overdutied:

None

14 Stability and Reactive Power

(Summary of the VAR requirements based upon the results of the dynamic studies)

To be determined in the Facilities Study Phase.

15 Affected Systems

None

16 Attachment 1: One Line Diagram

